1 State-Space Signal Model

A linear discrete-time state-variable system can be given as

$$\begin{aligned} \boldsymbol{\theta}[k] &= \mathbf{A}[k-1]\boldsymbol{\theta}[k-1] + \boldsymbol{\omega}[k] \qquad (1) \\ \mathbf{y}[k] &= \mathbf{S}[k]\boldsymbol{\theta}[k] + \boldsymbol{\psi}[k] \qquad (2) \end{aligned}$$

where $\boldsymbol{\theta}[k] \in \mathbb{R}^{P\times 1}$ is the parameter estimate at time instant k, $\mathbf{y}[k] \in \mathbb{R}^{O\times 1}$ is the observation data at time instant k, $\mathbf{A}[k] \in \mathbb{R}^{P\times P}$ is the state-space model, $\mathbf{S}[k] \in \mathbb{R}^{O\times P}$ is the observation (measurement) model, $\boldsymbol{\omega}[k] \in \mathbb{R}^{P\times 1}$ is the state noise process, and $\boldsymbol{\psi}[k] \in \mathbb{R}^{O\times 1}$ is the observation noise process.

Let us introduce the following assumptions:

1. The state noise process $\boldsymbol{\omega}[k]$ is zero mean with covariance

$$\mathrm{E}\left[\boldsymbol{\omega}[k]\boldsymbol{\omega}^{\mathrm{T}}[\ell]\right] = \mathbf{Q}[k]\,\delta_{k,\ell}. \qquad (3)$$

2. The observation noise process $\boldsymbol{\psi}[k]$ is zero mean with covariance

$$\mathrm{E}\left[\boldsymbol{\psi}[k]\boldsymbol{\psi}^{\mathrm{T}}[\ell]\right] = \mathbf{R}[k]\,\delta_{k,\ell}. \qquad (4)$$

3. The state and observation process noise have correlation

$$\mathrm{E}\left[\boldsymbol{\omega}[k]\boldsymbol{\psi}^{\mathrm{T}}[\ell]\right] = \mathbf{M}[k]\,\delta_{k,\ell}. \qquad (5)$$

4. There is some initial condition or a priori density, on the random variable $\boldsymbol{\theta}[0]$ with mean $\mu_{\boldsymbol{\theta}} = \mathrm{E}[\boldsymbol{\theta}[0]]$ and covariance $\mathrm{E}\left[\boldsymbol{\theta}[0]\boldsymbol{\theta}^{\mathrm{T}}[0]\right] = \boldsymbol{\Pi}$.

Thus, a Klaman filter to provide parameter estimates $\hat{\boldsymbol{\theta}}[k|k]$ considering all observations $\mathbf{y}[k]$ up to the time instant k can be defined with the following steps:

1. State estimate extrapolation (propagation):

$$\hat{\boldsymbol{\theta}}[k|k-1] = \mathbf{A}[k-1]\hat{\boldsymbol{\theta}}[k-1|k-1] \qquad (6)$$

2. Error covariance extrapolation (propagation):

$$\mathbf{P}[k|k-1] = \mathbf{A}[k-1]\mathbf{P}[k-1|k-1]\mathbf{A}^{\mathrm{T}}[k-1] + \mathbf{Q}[k-1] \qquad (7)$$

3. Kalman gain:

$$\mathbf{K}[k] = \mathbf{P}[k|k-1]\mathbf{S}^{\mathrm{T}}[k](\mathbf{S}[k]\mathbf{P}[k|k-1]\mathbf{S}^{\mathrm{T}}[k] + \mathbf{R}[k])^{-1} \tag{8}$$

4. State estimate update:

$$\hat{\boldsymbol{\theta}}[k|k] = \hat{\boldsymbol{\theta}}[k|k-1] + \mathbf{K}[k]\left(\mathbf{y}[k] - \mathbf{S}[k]\hat{\boldsymbol{\theta}}[k|k-1]\right) \tag{9}$$

5. Error covariance update:

$$\mathbf{P}[k|k] = (\mathbf{I} - \mathbf{K}[k]\mathbf{S}[k])\mathbf{P}[k|k-1] \tag{10}$$

2 State-Space Model

We assume that the delay and phase variations of the signal are due to the relative movement between the transmitter and the receiver, to synchronization mismatches, and to the propagation along the communication channel.

We approximate the channel by a sequence of time-variantly weighted taps equidistantly spaced in time.

$$h(t,\tau) \approx \sum_{l=0}^{L} h^{(l)}(t)\delta(\tau - lT_s) \tag{11}$$

where τ is the experienced delay, $h^{(l)}$ is the weight of the $l-th$ tap, $T_s = \dfrac{1}{2B}$ and L is large enough to satisfy:

- $LT_s = T_c$;
- $T_s \ll T_c$

Thus, the state vector for the sampled signal is

$$\boldsymbol{\theta}[k] = \begin{bmatrix} \tau[k] \\ \phi_\nu[k] \\ 2\pi\nu[k] \\ 2\pi a_\nu[k] \\ h^{(0)}[k] \\ h^{(1)}[k] \\ \vdots \\ h^{(L)}[k] \end{bmatrix} \in \mathbb{R}^{(L+5)\times 1}, \tag{12}$$

where

$$\phi_\nu[k] = \phi_\nu[k-1] + 2\pi T_d \nu[k-1] + 2\pi \frac{T_d^2}{2} a_\nu[k-1] \tag{13}$$

is the phase variation due to the movement between the transmitter and the receiver,

$$\tau[k] = \tau[k-1] + 2\pi T_d \beta \nu[k-1] + 2\pi \frac{T_d^2}{2} \beta a_\nu[k-1] \tag{14}$$

is the delay variation, whereas $a_\nu[k]$ is the angular Doppler drift, $\beta = \frac{1}{2\pi f_c}$, and f_c is the carrier frequency of the signal and

$$h^{(l)}[k] = \alpha^{(l)} h^{(l)}[k-1] \tag{15}$$

is the random walk dynamic model for the channel coefficients.

Including also noise, modeling random behavior, the state-space model can be given as

$$\boldsymbol{\theta}[k] = \mathbf{A}\,\boldsymbol{\theta}[k-1] + \boldsymbol{\omega}[k] \tag{16}$$

with

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & \beta T_d & \beta \frac{T_d^2}{2} & 0 & 0 & \ldots & 0 \\ 0 & 1 & T_d & \frac{T_d^2}{2} & 0 & 0 & \ldots & 0 \\ 0 & 0 & 1 & T_d & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 0 & 0 & \alpha^{(0)} & 0 & \ldots & 0 \\ 0 & 0 & 0 & 0 & 0 & \alpha^{(1)} & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \ldots & \alpha^{(L)} \end{bmatrix} \in \mathbb{R}^{(L+5)\times(L+5)}. \tag{17}$$

The state noise process is zero mean

$$\mathrm{E}\left[\boldsymbol{\omega}[k]\right] = \mathbf{0} \tag{18}$$

and has the covariance matrix

$$\mathrm{E}\left[\boldsymbol{\omega}[k]\boldsymbol{\omega}^{\mathrm{T}}[\ell]\right] = blkdiag\{\mathbf{Q_1}, \mathbf{Q_2}\} \tag{19}$$

where

$$\begin{aligned} \mathbf{Q_1} = \sigma_{clk_1}^2 & \begin{bmatrix} T_d\beta^2 & T_d\beta & 0 & 0 \\ T_d\beta & T_d & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \sigma_{clk_2}^2 \begin{bmatrix} \frac{T_d^3}{3}\beta^2 & \frac{T_d^3}{3}\beta & \frac{T_d^2}{2}\beta & 0 \\ \frac{T_d^3}{3}\beta & \frac{T_d^3}{3} & \frac{T_d^2}{2} & 0 \\ \frac{T_d^2}{2}\beta & \frac{T_d^2}{2} & T_d & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ & + (\sigma_{a_v}^2 + \sigma_{clk_3}^2) \begin{bmatrix} \frac{T_d^5}{20}\beta^2 & \frac{T_d^5}{20}\beta & \frac{T_d^4}{8}\beta & \frac{T_d^4}{6}\beta \\ \frac{T_d^5}{20}\beta & \frac{T_d^5}{20} & \frac{T_d^4}{8} & \frac{T_d^3}{6} \\ \frac{T_d^4}{8}\beta & \frac{T_d^4}{8} & \frac{T_d^3}{3} & \frac{T_d^2}{2} \\ \frac{T_d^3}{6}\beta & \frac{T_d^3}{6} & \frac{T_d^2}{2} & T_d \end{bmatrix} + \sigma_{iono}^2 \begin{bmatrix} T_d & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \tag{20}$$

where $\sigma_{a_v}^2$ is the variance of the angular Doppler error drift rate, $\sigma_{clk_1}^2$, $\sigma_{clk_2}^2$, and $\sigma_{clk_3}^2$ are modeling the drift behavior of the receiver clock, and σ^2_{iono} models the code-carrier divergence due to the ionospheric effect. The time-delay and Doppler estimate of the sate estimate extrapolation (propagation) is used for Doppler correction, as

$$\begin{bmatrix} \hat{\tau}[k|k-1] \\ \hat{\phi_\nu}[k|k-1] \\ \hat{\nu}[k|k-1] \\ \hat{a_\nu}[k|k-1] \\ \hat{h}^{(0)}[k|k-1] \\ \hat{h}^{(1)}[k|k-1] \\ \vdots \\ \hat{h}^{(L)}[k|k-1] \end{bmatrix} = \mathbf{A} \begin{bmatrix} \hat{\tau}[k-1|k-1] \\ \hat{\phi_\nu}[k-1|k-1] \\ \hat{\nu}[k-1|k-1] \\ \hat{a_\nu}[k-1|k-1] \\ \hat{h}^{(0)}[k-1|k-1] \\ \hat{h}^{(1)}[k-1|k-1] \\ \vdots \\ \hat{h}^{(L)}[k-1|k-1] \end{bmatrix}. \tag{21}$$

3 Measurement Model

Let the spreading sequence of one received satellite be

$$\mathbf{c}(\tau) = [c(\tau), \ldots, \ldots, c(nT_s - \tau), \ldots, c((N-1)T_s - \tau)]^{\mathrm{T}} \tag{22}$$

and the phase evolution caused by the Doppler be

$$\mathbf{d}[k, \nu[k], \phi_\nu[k]] = \left[\mathrm{e}^{j\phi_\nu[k]}, \ldots, \mathrm{e}^{j(\phi_\nu[k]+2\pi nT_s\nu[k])}, \ldots, \mathrm{e}^{j(\phi_\nu[k]+2\pi(N-1)T_s\nu[k])}\right]^{\mathrm{T}} \tag{23}$$

which will be simplified as

$$\mathbf{d}[k] = \left[\mathrm{e}^{j\phi_\nu[k]}, \ldots, \mathrm{e}^{j(\phi_\nu[k]+2\pi nT_s\nu[k])}, \ldots, \mathrm{e}^{j(\phi_\nu[k]+2\pi(N-1)T_s\nu[k])}\right]^{\mathrm{T}} \tag{24}$$

and

$$\hat{\mathbf{d}}[k] = \left[\mathrm{e}^{j\hat{\phi_\nu}[k|k-1]}, \ldots, \mathrm{e}^{j(\hat{\phi_\nu}[k|k-1]+2\pi nT_s\hat{\nu}[k|k-1])}, \ldots, \mathrm{e}^{j(\hat{\phi_\nu}[k|k-1]+2\pi(N-1)T_s\hat{\nu}[k|k-1])}\right]^{\mathrm{T}} \tag{25}$$

To estimate the time-delay, we employ a bank of symmetrical and equidistant correlators, applying Doppler and Doppler phase correction to all of them, around the prompt correlator, assumed to be sufficiently synchronized with the LOS signal, as follows

$$\mathbf{B}(\hat{\tau}[k|k-1],\hat{\nu}[k|k-1],\hat{\phi}_\nu[k|k-1]]) \triangleq \\ \left[\ \mathbf{c}(\hat{\tau}[k|k-1]-L\cdot\Delta)\odot\hat{\mathbf{d}}[k]\ldots\mathbf{c}(\hat{\tau}[k|k-1]+0)\odot\hat{\mathbf{d}}[k]\ldots\mathbf{c}(\hat{\tau}[k|k-1]+L\cdot\Delta)\odot\hat{\mathbf{d}}[k]\ \right] \in \mathbb{R}^{N\times(2L+1)} \tag{26}$$

where the spacing used is $\Delta = \frac{T_c}{L}$ and the delay estimate is within one chip around the signal delay, that is $|\hat{\tau}[k|k-1]-\tau[k]| \leq T_c/2$.

Therefore, the correlators are

$$\mathbf{y}[k] = \frac{1}{N}\left(\mathbf{x}^{\mathrm{H}}[k]\mathbf{B}(\hat{\tau}[k|k-1],\hat{\nu}[k|k-1],\hat{\phi}_\nu[k|k-1]])\right)^T \tag{27}$$

with the received signal

$$\mathbf{x}[k] = \sqrt{P[k]}g[k]\mathrm{e}^{j\phi[k]}\left(\mathbf{H}[k]\mathbf{c}(\tau[k])\right)\odot\left(\mathbf{d}[k,\nu[k],\phi_\nu[k]]\right)+\mathbf{n}[k], \tag{28}$$

where the noise is

$$\mathbf{n}[k] = [n(kNT_s),\ldots,n((kN+n)T_s),\ldots,n((kN+N-1)T_s)]^{\mathrm{T}}. \tag{29}$$

Furthermore, $T_s = \frac{1}{2B}$ is the sampling duration, $g[k] \in \{-1,1\}$ is the navigation message modulated onto the spreading sequence, $\phi[k]$ is the phase due to signal propagation, and $P[k]$ is the received power of the satellite signal.

The measurement sensibility matrix is the linearization of the measurement function around the a priori state vector estimate

$$\mathbf{S}[k] = \frac{\partial \mathbf{y}[k]}{\partial \boldsymbol{\theta}[k]}\Big|_{\boldsymbol{\theta}[k]=\boldsymbol{\theta}^{-}[k]}$$

$$= \begin{pmatrix} \frac{\partial y[k]}{\partial \tau[k]} \\ \frac{\partial y[k]}{\partial \phi_v[k]} \\ \frac{\partial y[k]}{\partial v[k]} \\ \frac{\partial y[k]}{\partial a_v[k]} \\ \frac{\partial y[k]}{\partial h^{(0)}[k]} \\ \frac{\partial y[k]}{\partial h^{(1)}[k]} \\ \vdots \\ \frac{\partial y[k]}{\partial h^{(L)}[k]} \end{pmatrix}\Big|_{\boldsymbol{\theta}[k]=\boldsymbol{\theta}[k]} \tag{30}$$

Let

$$\frac{\sqrt{P[k]}}{N} g[k]\mathrm{e}^{-j\phi[k]} = \gamma \text{ and } \mathbf{B}(\hat{\tau}[k|k-1], \hat{v}[k|k-1], \hat{\phi}_v[k|k-1]]) = \mathbf{B}[\mathbf{k}|\mathbf{k-1}] \tag{31}$$

so for the expansion of the correlator formula with the signal model, we have

$$\mathbf{y}[k] = \gamma\left(\left(\mathbf{c}^{\mathrm{H}}(\tau[k])\mathbf{H}^{\mathrm{H}}[k]\right) \odot \left(\mathbf{d}^{\mathrm{H}}[k]\right)\right)\mathbf{B}[k|k-1] + \frac{1}{N}\mathbf{n}^{\mathrm{H}}[k]\mathbf{B}[k|k-1] \tag{32}$$

Since $\mathbf{H}[k]$ is the sum of circulant matrices weighted by channel coefficients, the result of its multiplication with the code sequence is a vector circularly shifted in time. Therefore,

$$\mathbf{c}^{\mathrm{H}}(\tau[k])\mathbf{H}^{\mathrm{H}}[k] = \sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} c(\tau[k] + lT_s) \\ \vdots \\ c(nT_s - \tau[k] + lT_s) \\ \vdots \\ c((N-1)T_s - \tau[k] + lT_s) \end{bmatrix}^T \tag{33}$$

Developing the Hamadard product with the Doppler correction, we have

$$\sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} c(\tau[k]+lT_s)\mathrm{e}^{-j\phi_\nu[k]} \\ \vdots \\ c(nT_s-\tau[k]+lT_s)\mathrm{e}^{-j(\phi_\nu[k]+2\pi nT_s\nu[k])} \\ \vdots \\ c((N-1)T_s-\tau[k]+lT_s)\mathrm{e}^{-j(\phi_\nu[k]+2\pi(N-1)T_s\nu[k])} \end{bmatrix}^T \tag{34}$$

The result of the matrix multiplication with the correlator bank, for each line, is the vector multiplication

$$\sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} c(\tau[k]+lT_s)\mathrm{e}^{-j\phi_\nu[k]} \\ \vdots \\ c(nT_s-\tau[k]+lT_s)\mathrm{e}^{-j(\phi_\nu[k]+2\pi nT_s\nu[k])} \\ \vdots \\ c((N-1)T_s-\tau[k]+lT_s)\mathrm{e}^{-j(\phi_\nu[k]+2\pi(N-1)T_s\nu[k])} \end{bmatrix}^T \cdot \tag{35}$$

$$\begin{bmatrix} c(\hat{\tau}[k|k-1]+q\Delta)\mathrm{e}^{j\hat{\phi}_\nu[k]} \\ \vdots \\ c(nT_s-\hat{\tau}[k|k-1]+q\Delta)\mathrm{e}^{j(\hat{\phi}_\nu[k]+2\pi nT_s\hat{\nu}[k])} \\ \vdots \\ c((N-1)T_s-\hat{\tau}[k|k-1]+q\Delta)\mathrm{e}^{j(\hat{\phi}_\nu[k]+2\pi(N-1)T_s\hat{\nu}[k])} \end{bmatrix} \tag{36}$$

For simplification, let

$$\varepsilon_{\phi_\nu}^k = \phi_\nu[k]-\hat{\phi}_\nu[k|k-1] \text{ and } \varepsilon_\nu^k = \nu[k]-\hat{\nu}[k|k-1] \tag{37}$$

Therefore, we may write

$$\sum_{l=0}^{L} h^{*(l)}[k] \sum_{n=0}^{N-1} c(nT_s-\tau[k]+lT_s)c(nT_s-\hat{\tau}[k|k-1]+q\Delta)\mathrm{e}^{-j\varepsilon_{\phi_\nu}^k+2\pi nT_s\varepsilon_\nu^k} \tag{38}$$

for each element of the resulting vector.

In Iliapoulis, there is the following consideration

$$\sum_{n=0}^{N-1} c(nT_s-\tau[k]+lT_s)c(nT_s-\hat{\tau}[k|k-1]+q\Delta) = \int_T s(t-\tau[k]+lT_S)s(t-\hat{\tau}[k|k-1]+q\Delta)dt \tag{39}$$

We will also consider this approximation of the sum to an integral. Thus,

$$\sum_{n=0}^{N-1} c(nT_s - \tau[k] + lT_s)c(nT_s - \hat{\tau}[k|k-1] + q\Delta)\mathrm{e}^{-j2\pi nT_s\varepsilon_\nu^k} = \chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k) \tag{40}$$

$$\varepsilon_\tau^k = \tau[k] - \hat{\tau}[k|k-1] \tag{41}$$

where $\chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k)$ is the ambiguity function defined as

$$\chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k) = \int_T s(t - \tau[k] + lT_S)s(t - \hat{\tau}[k|k-1] + q\Delta)\mathrm{e}^{-j2\pi t\varepsilon_\nu^k} \tag{42}$$

$$y[k] = \gamma\mathrm{e}^{-j\varepsilon_{\phi\nu}^k} \sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} \chi(\varepsilon_\tau^k + lT_s - L\Delta, \varepsilon_\nu^k) \\ \vdots \\ \chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k) \\ \vdots \\ \chi(\varepsilon_\tau^k + lT_s + L\Delta, \varepsilon_\nu^k) \end{bmatrix}^T \tag{43}$$

3.1 Delay derivative

The elements dependent on the delay are the integral contained in the vector. Therefore,

$$\frac{\partial y[k]}{\partial \tau[k]} = \gamma\mathrm{e}^{-j\varepsilon_{\phi\nu}^k} \sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} \frac{\partial}{\partial\tau[k]}\chi(\varepsilon_\tau^k + lT_s - L\Delta, \varepsilon_\nu^k) \\ \vdots \\ \frac{\partial}{\partial\tau[k]}\chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k) \\ \vdots \\ \frac{\partial}{\partial\tau[k]}\chi(\varepsilon_\tau^k + lT_s + L\Delta, \varepsilon_\nu^k) \end{bmatrix}^T \tag{44}$$

3.2 Doppler phase derivative

The Doppler phase is only within the exponential term, so the derivative is

$$\frac{\partial y[k]}{\partial \phi_\nu} = -j\gamma \mathrm{e}^{-j\varepsilon^k_{\phi_\nu}} \sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} \chi(\varepsilon^k_\tau + lT_s - L\Delta, \varepsilon^k_\nu) \\ \vdots \\ \chi(\varepsilon^k_\tau + lT_s - q\Delta, \varepsilon^k_\nu) \\ \vdots \\ \chi(\varepsilon^k_\tau + lT_s + L\Delta, \varepsilon^k_\nu) \end{bmatrix}^T \tag{45}$$

3.3 Doppler shift derivative

$$\frac{\partial y[k]}{\partial \nu[k]} = \gamma \mathrm{e}^{-j\varepsilon^k_{\phi_\nu}} \sum_{l=0}^{L} h^{*(l)}[k] \begin{bmatrix} \frac{\partial}{\partial \nu[k]}\chi(\varepsilon^k_\tau + lT_s - L\Delta, \varepsilon^k_\nu) \\ \vdots \\ \frac{\partial}{\partial \nu[k]}\chi(\varepsilon^k_\tau + lT_s - q\Delta, \varepsilon^k_\nu) \\ \vdots \\ \frac{\partial}{\partial \nu[k]}\chi(\varepsilon^k_\tau + lT_s + L\Delta, \varepsilon^k_\nu) \end{bmatrix}^T \tag{46}$$

3.4 Angular Doppler drift derivative

None of the terms are dependent on the Angular Doppler drift, so the derivative is 0.

3.5 Channel coefficients derivative

Since there is a sum of vectors weighted by the channel coefficients, the derivative in respect to a given wheight is the corresponding vector, as in

$$\frac{\partial y[k]}{\partial h^{*(l)}[k]} = \gamma \mathrm{e}^{-j\varepsilon^k_{\phi_\nu}} \begin{bmatrix} \chi(\varepsilon^k_\tau + lT_s - L\Delta, \varepsilon^k_\nu) \\ \vdots \\ \chi(\varepsilon^k_\tau + lT_s - q\Delta, \varepsilon^k_\nu) \\ \vdots \\ \chi(\varepsilon^k_\tau + lT_s + L\Delta, \varepsilon^k_\nu) \end{bmatrix}^T \tag{47}$$

4 Development for numeric method calculations

The spread sequence signal can be written as the reverse fourier integral of its counterpart in frequency. This can be written as follows:

$$s(t) = \int_{-\infty}^{\infty} \sqrt{T_c} sinc(fT_c) \mathrm{e}^{j2\pi ft} df \tag{48}$$

However, given the properties of the fourier transform, the following equations are also valid:

$$s(t)\mathrm{e}^{-j2\pi\nu t} = \int_{-\infty}^{\infty} \sqrt{T_c} sinc((f_1+\nu)T_c) \mathrm{e}^{j2\pi f_1 t} df_1 \tag{49}$$

$$s(t+\tau) = \int_{-\infty}^{\infty} \sqrt{T_c} sinc(f_2 T_c) \mathrm{e}^{j2\pi f_2(t+\tau)} df_2 \tag{50}$$

Thus, the ambiguity function can be rewritten as a triple integral:

$$\chi(\varepsilon_\tau^k + lT_s - q\Delta, \varepsilon_\nu^k) = \int_T s(t-\tau[k]+lT_S)s(t-\hat{\tau}[k|k-1]+q\Delta)\mathrm{e}^{-j2\pi t\varepsilon_\nu^k}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sqrt{T_c} sinc((f_1+\varepsilon_\nu^k)T_c)\mathrm{e}^{j2\pi f_1 t}\sqrt{T_c} sinc(f_2T_c)\mathrm{e}^{j2\pi f_2(t+\varepsilon_\tau^k+lT_s-q\Delta)}dt\,df_1\,df_2$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} T_c sinc((f_1+\varepsilon_\nu^k)T_c) sinc(f_2T_c)\mathrm{e}^{j2\pi(f_1+f_2)t}\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}dt\,df_1\,df_2$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} T_c sinc((f_1+\varepsilon_\nu^k)T_c) sinc(f_2T_c)\delta(f_1+f_2)\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_1\,df_2$$

Where $\delta(x)$ represents Dirac's Delta Function. We can, without loss of generality, integrate first in the variable f_1:

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} T_c sinc((f_1+\varepsilon_\nu^k)T_c)sinc(f_2T_c)\delta(f_1+f_2)\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_1df_2$$

$$= \int_{-\infty}^{\infty} T_c sinc((-f_2+\varepsilon_\nu^k)T_c)sinc(f_2T_c)\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_2$$

Let B represent the signal's bandwith, then, with the new integration limits, the ambiguity integral becomes:

$$\chi(\varepsilon_\tau^k+lT_s-q\Delta,\varepsilon_\nu^k) = \int_{-B/2}^{B/2} T_c sinc((-f_2+\varepsilon_\nu^k)T_c)sinc(f_2T_c)\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_2 \tag{51}$$

This integral can be implemented for computation by numeric means.

4.1 Delay derivative

The following is true for the delay error derivative:

$$\frac{\partial}{\partial\tau[k]}\varepsilon_\tau^k = \frac{\partial}{\partial\tau[k]}(\tau[k]-\hat{\tau}[k|k-1]) = 1 \tag{52}$$

Given 51, the delay derivative can be foun evaluating the integral:

$$\begin{aligned}\frac{\partial}{\partial\tau[k]}\chi(\varepsilon_\tau^k+lT_s-L\Delta,\varepsilon_\nu^k) &= \frac{\partial}{\partial\tau[k]}\int_{-B/2}^{B/2} T_c sinc((-f_2+\varepsilon_\nu^k)T_c)sinc(f_2T_c)\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_2\\ &= \int_{-B/2}^{B/2} T_c sinc((-f_2+\varepsilon_\nu^k)T_c)sinc(f_2T_c)\frac{\partial}{\partial\tau[k]}\left(\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}\right)df_2\\ &= j2\pi\int_{-B/2}^{B/2} T_c sinc((-f_2+\varepsilon_\nu^k)T_c)sinc(f_2T_c)f_2\mathrm{e}^{j2\pi f_2(\varepsilon_\tau^k+lT_s-q\Delta)}df_2\end{aligned}$$

4.2 Doppler shift derivative

Similarly, for the Doppler shift error derivative:

$$\frac{\partial}{\partial \nu[k]} \varepsilon_{\nu}^{k} = \frac{\partial}{\partial \nu[k]} (\nu[k] - \hat{\nu}[k|k-1]) = 1 \tag{53}$$

$$\begin{aligned}
\frac{\partial}{\partial \nu[k]} \chi(\varepsilon_{\tau}^{k} + lT_s - L\Delta, \varepsilon_{\nu}^{k}) &= \frac{\partial}{\partial \nu[k]} \int_{-B/2}^{B/2} T_c sinc((-f_2 + \varepsilon_{\nu}^{k})T_c) sinc(f_2 T_c) \mathrm{e}^{j2\pi f_2(\varepsilon_{\tau}^{k} + lT_s - q\Delta)} df_2 \\
&= \int_{-B/2}^{B/2} T_c \frac{\partial}{\partial \nu[k]} \left(sinc((-f_2 + \varepsilon_{\nu}^{k})T_c) \right) sinc(f_2 T_c) \mathrm{e}^{j2\pi f_2(\varepsilon_{\tau}^{k} + lT_s - q\Delta)} df_2 \\
&= \int_{-B/2}^{B/2} T_c \left[sinc'((-f_2 + \varepsilon_{\nu}^{k})T_c) \cdot T_c \right] sinc(f_2 T_c) f_2 \mathrm{e}^{j2\pi f_2(\varepsilon_{\tau}^{k} + lT_s - q\Delta)} df_2 \\
&= \int_{-B/2}^{B/2} T_c^2 sinc'((-f_2 + \varepsilon_{\nu}^{k})T_c) sinc(f_2 T_c) f_2 \mathrm{e}^{j2\pi f_2(\varepsilon_{\tau}^{k} + lT_s - q\Delta)} df_2
\end{aligned}$$

Note from eq. (29) of "Iliopoulos, A., Enneking, C., Crespillo, O. G., Jost, T., Thoelert, S., & Antreich, F. (2017). Multicorrelator signal tracking and signal quality monitoring for GNSS with extended Kalman filter. 2017 IEEE Aerospace Conference, 1–10. https://doi.org/10.1109/AERO.2017.7943579" that it is necessary to calculate a predicted measurement \mathbf{z}^-[k] (which can also be written as \mathbf{z}[k \ mid k - 1]) in order to use the EKF algorithm. In other words, you need to 'recreate' the measurements vector y[k] from your formulation (which could be represented as \mathbf{z}[k] as i've mentioned earlier) using the states vector \hat{x}[k \mid k - 1].

Observe that in your theoretical framework you propose to **correct the Doppler effect and the time-delay together with the multicorrelator bank**. Therefore, i think that it would be necessary to apply a LQG controller to estimate the errors \varepsilon^{k}_{\tau} and \varepsilon^{k}_{\nu} in order to apply them on the measurements prediction, as it is proposed in Section 3 of "Fohlmeister, F. (2021). GNSS Carrier Phase Tracking under Ionospheric Scintillations [Thesis, Technische Universität München]. In Shaker Verlag. https://elib.dlr.de/185344/".

In my opinion, there is no gain in tracking the error state instead of the state itself. Therefore, i understand that we have two options at hand:

- To do not correct neither the Doppler effect nor the time-delay with the correlator bank. In this case, we would simply apply the correlator bank on the received signal \mathbf{y}[k] directly (without any correction). Then, on the EKF state space model, we would need to adjust the Jacobian matrix accordingly and we would need to apply both the Doppler effect and the time-delay corrections on the measurements prediction \mathbf{z}[k \mid k - 1].

- To do not track neither the Doppler effect nor the time-delay inside the EKF algorithm. This other option would have two Kalman filters working together: One fully dedicated for the channel impulse response modeling and the other one dedicated for the carrier-phase and time-delay correction. The main disadvantege of this approach is that the time-delay and the carrier-phase would not be preserved from the channel effects. However, it would be still possible to use the estimated channel coefficients in post-processing to correct channel effects.

I'll write a full report on both approaches that i'm proposing, so we can further discuss them.